

me after doing approximately
1 (one) productive thing

Day 11 - Help Session



Announcements

- HWs are getting graded
 - Submit Regrade request with evidence for concerns
- Project planning in starting with HW 4; starting thinking about a partner/group for final
- Office hours today 14:00-16:00 (BPS 1248)

HW 4 Exercise 3, Sliding puck

A small puck rests on a fixed sphere of radius R . The puck is given a tiny nudge and it slides down the sphere. Using conservation of energy, we can determine the point at which the puck leaves the sphere.

- 3a Setup the problem with a sketch. Explain the setup and include any assumptions that you need to make in order to solve the problem analytically. Identify the height as a function of the polar angle, $h(\theta)$. What is the maximum possible angle θ that the puck could reach before falling off? Why?

HW 4 Exercise 3, Sliding puck

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- 3b Use conservation of energy to find the speed of the puck as a function of its height. Your answer should be in terms of the polar angle, θ .
- 3c Use Newton's Second Law to find the normal force acting on the puck as a function of its height. Your answer should be in terms of the polar angle, θ . What is the condition for the puck to leave the sphere?

HW 4 Exercise 3, Sliding puck

A small puck rests on a fixed sphere of radius R . The puck is given a tiny nudge and it slides down the sphere. Using conservation of energy, we can determine the point at which the puck leaves the sphere.

- 3d At what angle and height does the puck leave the sphere?